Aerospace Science and Technology ••• (••••) •••–•••

Contents lists available at ScienceDirect

Aerospace Science and Technology

www.elsevier.com/locate/aescte

Distributed UAV flocking control based on homing pigeon hierarchical strategies

Qinan Luo, Haibin Duan

State Key Laboratory of Virtual Reality Technology and Systems, School of Automation Science and Electrical Engineering, Beihang University (formerly Beijing University of Aeronautics and Astronautics, BUAA), Beijing 100083, PR China

ARTICLE INFO

Article history:
Received 9 November 2016
Received in revised form 22 May 2017
Accepted 9 August 2017
Available online xxxx

Keywords:
Unmanned aerial vehicle (UAV)
Flocking control
Collective behavior
Pigeon hierarchical strategy

ABSTRACT

The recent boom of animal collective motion investigation has attracted many researchers to the field of applying such intelligence to complicated distribute control problems in artificial systems, such as autonomous unmanned aerial vehicle (UAV) flocking. In this paper, a distributed control framework inspired by homing pigeon hierarchical strategies is proposed to solve the UAV flocking problem. This new approach combines the advantages of velocity correlation, leader-follower interaction and hierarchical leadership network observed in pigeon flock with altitude consensus control algorithm used in UAV flocking. The flocking control algorithm is implemented to achieve a stable performance by controlling local position and velocity of each UAV. The practical dynamic and constraints of fix-wing UAV are also taken into account. The distributed flocking algorithms are tested in several simulation cases, which verify the effectiveness and applicability of the proposed control framework.

1. Introduction

Unmanned aerial vehicle (UAV) is becoming an integral part to possess the future world of artificial intelligence. UAV will be used for complex tasks including surveillance, agriculture irrigation, forestry fire prevention, cargo transportation and coordinated rescue missions in the presence of disturbances, failures, and uncertainties [1–5]. Much effort is currently spent on various control problems associated with UAVs moving in formation with limited information. The advantages of performing formation flight include fuel saving at certain close formation positions, efficiency improving in cooperative task allocation and robustness increasing in a self-organizing system [6–8]. In addition to the research on the formation flight of multi-UAV systems, there have been some studies on the coordinated control of UAV flock. Flocking is a spectacular phenomenon in nature. In a natural flock, a large number of similar individuals achieve complex and fascinating motions based on local autonomous interactions. Generally, flocking is more effective and robust in solving complex tasks such as foraging, long-distance migration and defend environmental threats. Such coordinate mechanisms in nature provide bio-inspiration for developing efficient flocking of UAVs with distributed control. Furthermore, flocking flight can extend existing advantages and exploit potential advantages of coordinated UAVs [9,10].

Over the last decade, rapid progress in sensor technology leads to many advances in research of free-flying bird orientation strategies, especially pigeon hierarchical strategies. The collective route of pigeon flock emerges as a compromise between individual's preferred path and the leader's experience. Nagy et al. [11] record the pigeons' movements during spontaneous and homing flights to investigate the temporal relationship among the pigeon individual and its neighbors. They also present a well-defined hierarchy of pigeon flock and the experimental results demonstrate that the hierarchical organization can increase the efficiency of group flight. Furthermore, in [12] Pettit et al. investigate the collective intelligence of pigeon flock in social learning by adding a locally naive individual as a flight partner. They also find that there exists a feedback loop between leadership and learning in collective motion over time [13]. Pigeon flock's organizational structure can be transformed into unique components of the distributed system, and the leadership hierarchies can be implemented as the control framework of UAV flock.

There are a series of animal flocking principles, being able to perform coordinated control for autonomous UAVs. For example, a decentralized multi-copter flock is presented by Vásárhelyi et al. based on bio-inspired strategies with limited dynamic information [14]. An autonomous flock control algorithm is proposed by Hauert et al. for fixed-wing UAVs by integrating two Reynolds rules into the control system [15]. A remarkable flock of 20 quadcopters with a central controller calculating the navigational information for all agents is created by Kushleyev et al. [16]. Another quadcopter flock

E-mail address: hbduan@buaa.edu.cn (H. Duan).

http://dx.doi.org/10.1016/j.ast.2017.08.010
1270-9638/

that is created by Bürkle et al. also implements the central processing at a ground station [17]. However, autonomous flock with decentralized control frameworks and self-organization capabilities needs to be further developed, and we take full advantages of animal flocking behaviors to achieve highly coherent motion.

Homing pigeons have been frequently used as a model in flocking strategies studies. Emerging evidence indicates that hierarchical leadership network dominates pigeon flock behavior, with most flock members following the decisions made by high level leaders in the network. Furthermore, in flocks composed of the same individuals the leadership hierarchy tends to be structurally stable. Pigeons may share navigational decisions and decide democratically on leadership when there is little conflict of opinion about the route to follow. These homing pigeon leadership hierarchies can be applied to solve complicated distribute control problems in UAV flocking.

In this paper, a distributed control framework is proposed to solve the UAV flocking problem. This framework contains various hierarchical strategies based on the pairwise leader-follower relations in pigeon flock and condenses those interactions into robust, fully transitive leadership hierarchies. This framework can be used to construct a stable flocking structure, and it can contain as many system-specific features as we can take into account, such as inertia, time delay, general noise and inner noise. The hierarchical features of pigeon flock can be useful in collective UAVs only if some specific aspects of realistic systems are considered. The principles of pigeon flock hierarchy can be transformed into unique control patterns of the UAV dynamical system. With simulations and further experiments on UAV flocking, the stability and efficiency of pigeon-inspired flocking control framework are studied.

The rest of this paper is organized as follows. Section 2 introduces the hierarchical strategies in pigeon flock, including velocity correlation, leader-follower interaction, and hierarchical leadership. Section 3 gives a description of the fix-wing UAV model and the transformation of control inputs. In Section 4, the proposed distributed control framework based on pigeon leadership hierarchies is implemented on the fix-wing UAV flocking. Series of simulation results and relevant analysis are discussed in Section 5, followed by our concluding remarks in Section 6.

2. Homing pigeon leadership hierarchies

This section presents three typical hierarchical strategies in pigeon flock, including velocity correlation, leader-follower interaction and hierarchical leadership network. Direction and velocity choice dynamics within flocks of up to 8 homing pigeons, which are obtained by high-resolution lightweight GPS devices [18], are considered. The relationship between individual pigeon's velocity performance during free flights and homing flights are examined. In general, considering about a pigeon flock of N individuals, each individual is represented by the position vector $p_i(t)$ and the velocity vector $v_i(t)$. The projected distance $d_{ij}(t)$ is defined as the distance between pigeons i and j on the motion direction of the whole flock. The velocity correlation function $C_{ij}(\tau)$ can be defined as:

$$C_{ij}(\tau) = \langle \boldsymbol{v}_i(t) \cdot \boldsymbol{v}_j(t+\tau) \rangle \tag{1}$$

where $\langle \bullet \rangle$ denotes the time average, τ represents the time delay of velocity correlation behavior between pigeons i and j. The directional correlation delay time is defined as τ_{ij}^*, and the maximum value of velocity correlation function $C_{ij}(\tau)$ is $C_{ij}(\tau_{ij}^*)$. The directional correlation delay time represents the strength of pairwise leader-follower relationship between individuals. The velocity correlation behavior can be considered as a basic "velocity copy" motion related to pigeon flock hierarchical leadership. A pigeon is defined as a leader when its velocity is 'copied' by another pigeon delayed in time. About two-thirds of pairwise comparisons between pigeons produced clearly directed velocity correlation edges, and the average delay time of this behavior is 0.3 s. The characteristic delay time of each individual can be presented as pigeons' hierarchical reaction performance in the context of following a persistent change in the direction of motion of leaders.

According to the velocity correlation analysis, pigeons tend to copy the direction and velocity of particular individuals (neighbors) consistently. In order to examine the effects of pigeon hierarchical organization on stability and robustness, we extend an existing collective motion model [19] by further incorporating the pigeon inspired leader-follower interaction and hierarchical leadership network. In this model, individuals interact with their neighbors within a fixed communicating range, these neighbors should be higher leaders in the flock. We divide the communicating area into three specific interaction zones: the avoidance zone (r_R), the alignment zone (r_O) and the attraction zone (r_A). At all times, each pigeon tries to maintain a minimum distance between itself and other pigeons by turning away from individuals within the avoidance zone. The preferred turning direction of pigeon i can be expressed as:

$$\psi_i(t+\Delta t) = -\sum_{j \neq i} \frac{p_j(t) - p_i(t)}{|p_j(t) - p_i(t)|} \tag{2}$$

where $\psi_i(t+\Delta t)$ is the individual's desired direction of travel at $t+\Delta t$. The leader-follower interaction in pigeon flock mainly occurs within the attraction zone and the alignment zone. Individuals are attracted to and align with their leaders based on the velocity correlation. Then the turning direction can be calculated as:

$$\psi_i(t+\Delta t) = \sum_{j \neq i} \frac{p_j(t) - p_i(t)}{|p_j(t) - p_i(t)|} + \sum_{j=1}^{N} \frac{v_j(t)}{|v_j(t)|} \tag{3}$$

While navigating towards a fixed target location, leaders must balance their preference to maintain flock cohesion with their global navigation behavior. Then the direction can be calculated as:

$$\psi_i'(t+\Delta t) = \frac{(1-\omega)\psi_i'(t+\Delta t) + \omega\psi_{ig}}{|(1-\omega)\psi_i'(t+\Delta t) + \omega\psi_{ig}|} \tag{4}$$

where $\psi_i'(t+\Delta t)$ represents the leader's preferred direction at $t+\Delta t$. ω is a weighting factor ranges from 0 to 1. $\omega = 0$ implies no navigation towards the preferred direction and $\omega = 1$ represents only the use of navigational information and no social interactions. In order to select a suitable value of weighting factor ω, series of simulations are implemented on the basis of the global navigation behavior with ω varying from 0 to 1. A higher value of ω is better in the premise of maintaining the integrality of pigeon flock. ψ_{ig} is the direction towards the global target. The factor represents the balance between leader's social interaction and its navigation towards the target.

Crucially, the velocity correlation and leader-follower interaction in pigeon flock are based on a robust hierarchical leadership network, containing only directed transitive leader-follower relationships [20,21]. The hierarchical leadership network is divided into several levels, and pigeons in higher hierarchy level are more influential in determining the flock's movement. Furthermore, leaders in higher level of the network have more followers. Therefore, the first level leader can contribute with relatively most weight to the movement decisions of the flock, as it has most followers who consistently copy its movement. With multiple leaders in the second and the third levels of the hierarchically organized network, pairwise leader-follower relations among flock members remain stable in emergency situations. Leadership ranks within

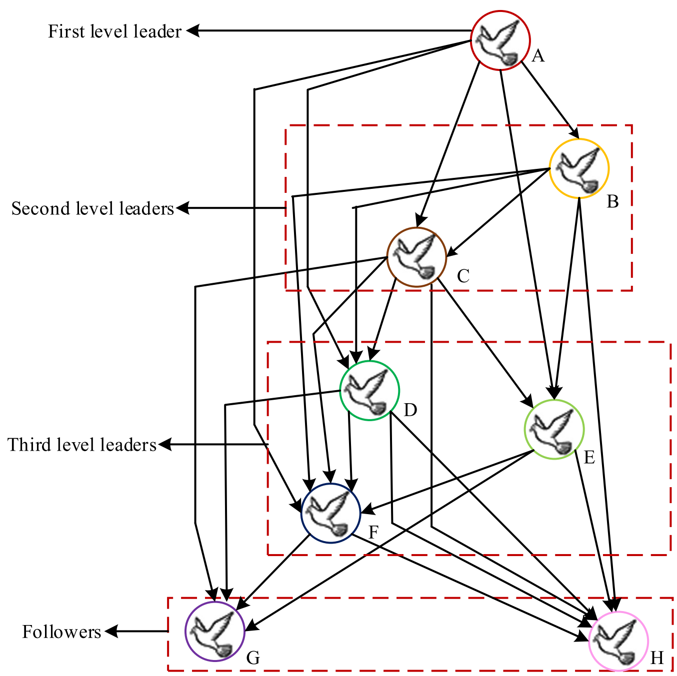


Fig. 1. Homing pigeon hierarchical leadership network.

flocks relate to a combination of several properties, such as navigation experience, flight speed and physical strength.

The hierarchical leadership network demands less attention from group members, and it is resistant to disturbances. The velocity correlation and leader-follower dynamics are embedded in the hierarchical network, which results in more efficient information transfer and coordination among group members. These hierarchical strategies in pigeon flock inspire us to design a distributed control framework to solve the UAV flocking problem. In order to reduce the computation complexity on the UAV communicating network, we remove the edges from the first level leader to the followers at the bottom (shown in Fig. 1). Each UAV can interact with its flock mates in a way that is defined by the pigeon hierarchical features, which include a consequence of distributed interaction rules. The hierarchical network mechanisms can allow the establishment and maintenance of robust UAV flocking.

3. Fix-wing UAV model

This paper mainly focuses on the high-level flocking control design of UAVs. We assume that each UAV is equipped with airspeed, heading, and altitude autopilots. Then the 12-dimensional full state model of the UAV can be simplified to a representative point-mass model [22,23] as shown in Fig. 2. Consider a flock of M UAVs with the following model:

$$\begin{aligned}
\dot{x}_i &= V_i \cos\gamma_i \cos\chi_i \\
\dot{y}_i &= V_i \cos\gamma_i \sin\chi_i \\
\dot{h}_i &= V_i \sin\gamma_i \\
\dot{V}_i &= \frac{T_i - D_i}{m_i} - g\sin\gamma_i \\
\dot{\gamma}_i &= \frac{L_i\cos\phi_i - m_i g\cos\gamma_i}{m_i V_i} \\
\dot{\chi}_i &= \frac{L_i \sin\phi_i}{m_i V_i \cos\gamma_i}, \quad i = 1, \ldots, M
\end{aligned} \tag{5}$$

where x_i is the displacement in the forward direction, y_i is the displacement in the lateral direction, h_i is the altitude, V_i is the ground speed which is assumed to be equal to the airspeed without any disturbance. γ_i, ϕ_i, and χ_i are the flight path angle, the banking angle, and the heading angle, respectively. T_i, D_i, and L_i represent the thrust, the drag, and the lift, respectively. m_i is the mass and g is the gravitational acceleration.

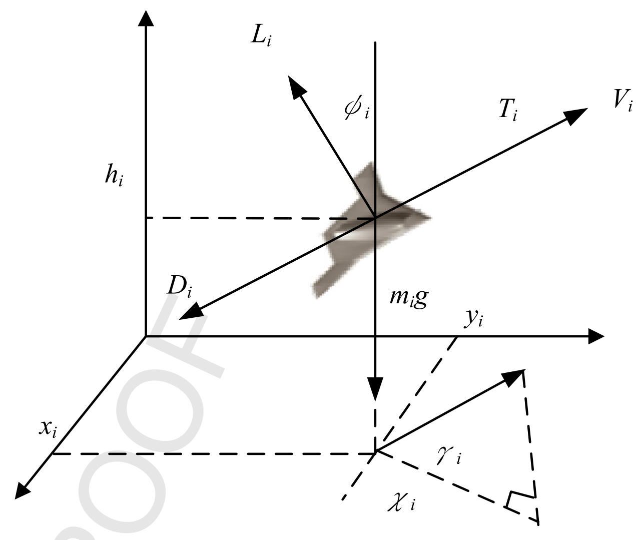


Fig. 2. Reference frames of the UAV model.

The nonlinear UAV model can be pre-linearized using feedback linearization, which is presented as:

$$\begin{aligned}
\ddot{x}_i &= u_{x_i} \\
\ddot{y}_i &= u_{y_i} \\
\ddot{h}_i &= u_{h_i}
\end{aligned} \tag{6}$$

where u_{x_i}, u_{y_i}, and u_{h_i} are virtual acceleration control inputs related to the plane movements and altitude changes. The banking angle, lift, and thrust are chosen as real control inputs, which can be calculated as:

$$\begin{aligned}
\phi_i &= \tan^{-1}\left(\frac{u_{y_i}\cos\chi_i - u_{x_i}\sin\chi_i}{(u_{h_i}+g)\cos\gamma_i - (u_{x_i}\cos\chi_i + u_{y_i}\sin\chi_i)\sin\gamma_i}\right) \\
L_i &= m_i \frac{(u_{h_i}+g)\cos\gamma_i - (u_{x_i}\cos\chi_i + u_{y_i}\sin\chi_i)\sin\gamma_i}{\cos\phi_i} \\
T_i &= m_i\left[(u_{h_i}+g)\sin\gamma_i + (u_{x_i}\cos\chi_i + u_{y_i}\sin\chi_i)\cos\gamma_i\right] + D_i
\end{aligned} \tag{7}$$

Additionally, V_i, χ_i, and $\dot{h}_i$ should satisfy the practical dynamic constraints of the fixed-wing UAV:

$$\begin{aligned}
&V_{\min} \le V_i \le V_{\max} \\
&|\chi_i| \le n_{\max} g / V_i \\
&\lambda_{\min} \le \dot{h}_i \le \lambda_{\max}
\end{aligned} \tag{8}$$

where $V_{\min}$, $V_{\max}$, $n_{\max}$, $\lambda_{\min}$, and $\lambda_{\max}$ are constants corresponding to the minimum velocity, the maximum velocity, the maximum lateral overload, the maximum gliding speed, and the maximum climbing speed, respectively.

4. Pigeon leadership hierarchies based distributed control framework

In this section, a distributed framework of the flocking control is proposed for the UAV flock trajectory tracking based on pigeon leadership hierarchies and altitude consensus. As illustrated in Fig. 3, in order to allow the UAV flock to fly in the same horizontal plane, the control framework is decoupled into two parts: the horizontal control and vertical control. The vertical control

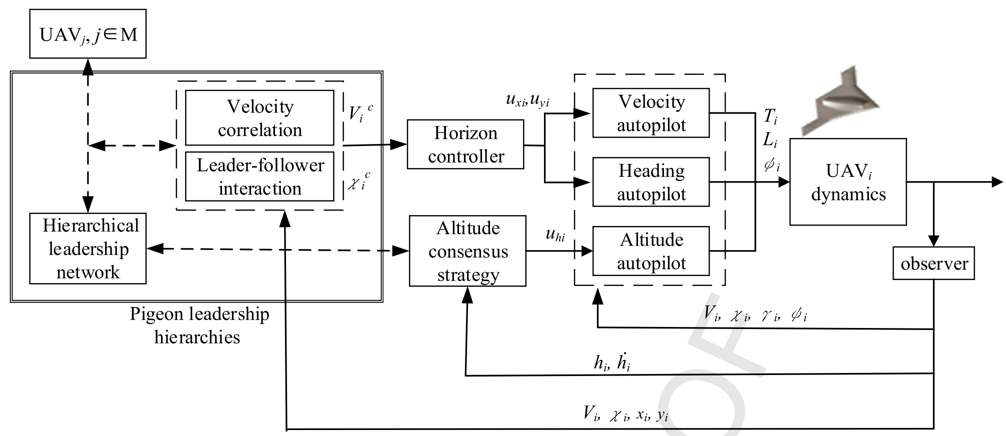


Fig. 3. The block diagram of the flocking control system based on pigeon leadership hierarchies.

module uses the altitude consensus strategy to calculate the control inputs of altitude autopilot, which guarantees that all UAVs fly in the same horizontal plane. The horizontal control module is established based on the distributed flocking control algorithms inspired by the homing pigeon, which provides control inputs to the velocity and heading autopilots. The horizontal control module includes velocity correlation, leader-follower interaction, and hierarchical leadership network. All the control modules allow the UAV flock to keep the flight formation and track the first level leader's trajectory.

The horizontal control is based on the pigeon leadership hierarchies. Each of the continuous control inputs for the ith UAV can be approximated by a piecewise function with constant as follows:

$$u_i = f_i^p + f_i^a + f_i^o \tag{9}$$

where f_i^p, f_i^a, and f_i^o represent velocity correlation, leader-follower interaction, and hierarchical leadership, respectively. The velocity correlation of the pigeon inspired control strategies can be written as follows:

$$\begin{aligned}
&f_i^p = \omega_{rep} f_{i,rep} + \omega_{att} f_{i,att} + \omega_{al} f_{i,al} + \omega_{front} f_{i,front} \\
&f_{i,rep} = \frac{1}{n_{rep}} \sum_{j=1}^{n_{rep}} g(d_{ij}) \vec{u}_{ij}, \quad 0 < d_{ij} < r_{rep} \\
&f_{i,att} = \frac{1}{n_{att}} \sum_{j=1}^{n_{att}} g(d_{ij}) \vec{u}_{ij}, \quad r_{al} < d_{ij} < r_{att} \\
&f_{i,al} = \frac{1}{n_{al}} \sum_{j=1}^{n_{al}} \frac{\vec{v}_j}{|\vec{v}_j|}, \quad r_{rep} < d_{ij} < r_{al} \\
&f_{i,front} = g_f(d_{ij}) \vec{u}_{ij}
\end{aligned} \tag{10}$$

where ω_{rep}, ω_{att}, ω_a, and ω_{front} represent weight coefficients of repulsive force, attractive force, alignment force, and forward propulsion. d_{ij} represents the relative distance between UAV i and j. $\vec{v}_j$ represents the velocity of the jth UAV. $\vec{u}_{ij}$ represents unit vector of the relative distances. n_{rep}, n_{att}, n_{al} are normalization factors. r_{rep}, r_{al}, and r_{att} represent radiuses of repulsive force, alignment force, and attractive force, respectively.

The leader-follower interaction, and hierarchical leadership strategies can be further expressed as follows:

$$f_i^a = \sum_{j \neq i} \frac{p_j(t) - p_i(t)}{|p_j(t) - p_i(t)|} + \sum_{j=1}^{N} h_{ij} \frac{v_j(t)}{|v_j(t)|} \tag{11}$$

$$\begin{aligned}
&f_i^o = \alpha_i(t) + P_i(t) + O_i(t) + A_i(t) + E_i(t) \\
&P_i(t) = \begin{cases} \lambda \psi_i(t) e^{-(\psi_i(t)/d_0)} \\ -\lambda \psi_i(t) e^{-(\psi_i(t)/d_0)} \end{cases} \\
&O_i(t) = a \tanh\big(\varphi_i(t) a_{sl}\big) \\
&A_i(t) = c \sin\big(\theta_i(t)\big) \tanh\big(\big(r_i(t) - r_0\big) r_{sl}\big)
\end{aligned} \tag{12}$$

where $\psi_i(t)$ and $p_i(t)$ represent the heading angle and the position of the ith UAV, respectively. λ, a_{sl}, a, and d_0 are control factors related to heading angle change rate.

The vertical control is based on an altitude consensus control strategy expressed as:

$$\begin{aligned}
u_{h_i} = &-c_{\dot{h}} \sum_{i,j \in M, i \neq j} (\dot{h}_i - \dot{h}_j) - c_h \sum_{i,j \in M, i \neq j} (h_i - h_j) \\
&- c_{\dot{h}}^1 (\dot{h}_i - \dot{h}_D) - c_h^1 (h_i - h_D)
\end{aligned} \tag{13}$$

where $c_{\dot{h}}$, c_h, $c_{\dot{h}}^1$, and c_h^1 are positive constants. This altitude consensus control strategy guarantee that $\dot{h}_i \to \dot{h}_D$ and $h_i \to h_L$. The horizontal flocking control is based on the distributed flocking control algorithm inspired by the pigeon leadership hierarchies. A transformation from the pigeon hierarchical strategies to the virtual control vector is constructed. The relationships between the virtual control vector $\boldsymbol{u}_i = [u_{x_i}, u_{y_i}, u_{h_i}]$ and the pigeon flock dynamics can be expressed as:

$$\begin{aligned}
&V_i^c = \tau_v (u_{x_i} \cos \chi_i + u_{y_i} \sin \chi_i) + V_i \\
&\chi_i^c = \frac{\tau_\chi}{V_i} (u_{y_i} \cos \chi_i - u_{x_i} \sin \chi_i) + \chi_i
\end{aligned} \tag{14}$$

where V_i^c and χ_i^c are the desired velocity and travel direction calculated based on the velocity correlation and the leader-follower interaction of pigeon flock. τ_v and τ_χ are positive time delay constants for the velocity and heading angle related to the pigeon hierarchical network. The time delay is added when a UAV receives and processes position and velocity data from other UAVs. The velocity alignment rule in the pigeon flock term in realistic control algorithms should consider the velocity difference of UAVs close to each other and it has an upper threshold value. Besides the practical dynamic constraints of the fixed-wing UAV, a viscous friction-like interaction is adopted in the alignment rule, which is given as:

$$v_{ij}^{frict} = C_{frict} \frac{v_j - v_i}{(\max\{r_{\min}, |d_{ij}|\})^2} \tag{15}$$

where C_{frict} is the strength of the friction and $r_{\min}$ represents a threshold to avoid division by close-to-zero distances. A Gaussian outer noise $\eta_i(t)$ with standard deviation σ is also added to

Table 1
Parameters of the flocking interactions.

Parameter	Notation	Value
Radius of attraction zone (m)	r_A	150
Radius of alignment zone (m)	r_O	20
Radius of avoidance zone (m)	r_R	3
Standard deviation of the Gaussian noise	σ	0.2
Minimum interaction threshold (m)	$r_{\min}$	1
Control gains of the altitude consensus algorithm	$[c_h, c_h, c_h^1, c_h^1]$	[1, 2, 1, 2]

Table 2
Model parameters of the UAV dynamics.

Parameter	Notation	Value
Velocity time constant	τ_v	0.3
Heading angle time constant	τ_χ	0.25
Minimum and maximum velocity (m/s)	$[V_{\min}, V_{\max}]$	[5, 15]
Maximum climbing and gliding velocity (m/s)	$[\lambda_{\min}, \lambda_{\max}]$	[−5, 5]
Maximum lateral overload (g)	$n_{\max}$	5

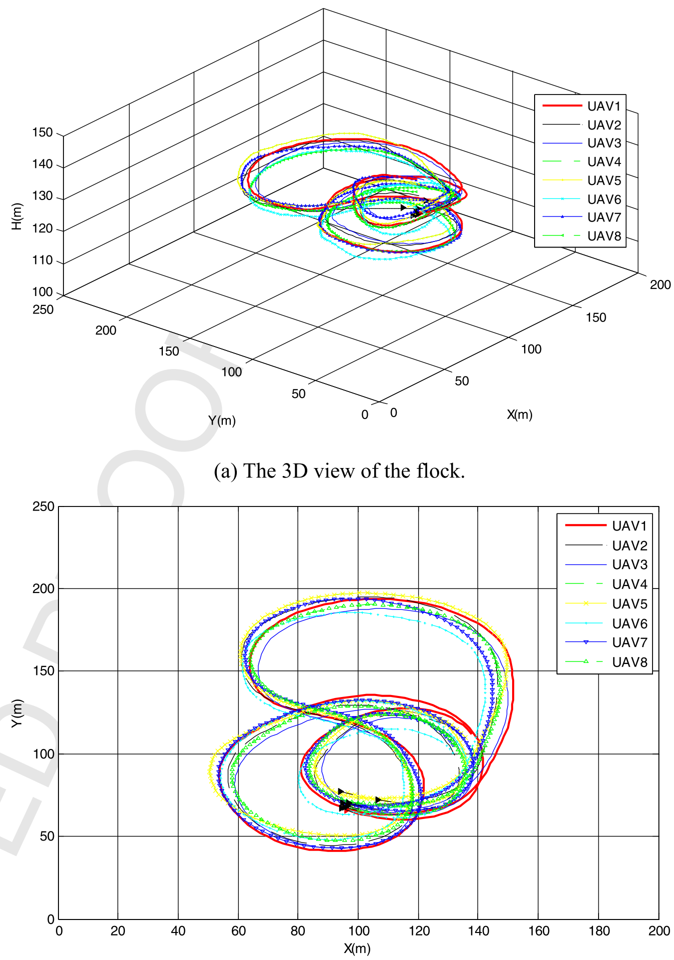
the velocity measurement of UAVs. This noise is a model of unpredictable environmental effects such as wind disturbances and the compensation of the low-level controller. To study the flocking control framework with simulations, the global positional constraint of the UAV flock should be considered, which can be well substituted by using periodic boundary conditions. In real experiments, periodic boundary conditions can be achieved by attracting the UAVs into a ring-shaped arena.

5. Simulation results and analysis

Series of simulations are implemented on the basis of the control framework inspired by the pigeon leadership hierarchies, which is described in the previous sections. Several simulation results of the 3D UAV flocking control are presented to verify the stability of the UAV flock, which can be guaranteed with the distributed control framework inspired by the pigeon leadership hierarchies. The hierarchical leadership based flocking control algorithm and the consensus based altitude control strategy are tested in two scenarios. The effects of the hierarchical network on the stability are investigated by introducing an emergency situation that a middle-level leader suddenly disappears. Control parameters are optimized to achieve a stable formation of the flocking with smooth maneuvering. Parameters of the flocking interactions are shown in Table 1. All the algorithms are implemented using a 3.07 GHz CPU and 6G memory personal computer running Windows 7 × 64 and Matlab R2012b. The step-size is set to be $\Delta t = 0.025$ s. The UAV flock in simulations consists of $M = 8$ fixed-wing UAVs corresponding to pigeons A–H in the hierarchical network. The model parameters of the UAV dynamics are shown in Table 2.

At the beginning of the simulation, UAVs are placed randomly within a 35 m wide square-shaped area. The desired trajectory of the UAV flock is a lemniscate with random variables, which means that the flock would fly within a specified area with some randomness. Fig. 4 shows the 3D simulation results of UAV flocking using the proposed control algorithms and the altitude consensus-based algorithm. In Fig. 4, the red point represents the first level leader (UAV1), which determines the trajectory of the flock. The red line represents the first level leader's trajectory, while the other lines represent the followers' trajectories. The arrow in the front of each line represents the direction of current motion.

The experimental results show that the proposed consensus-based altitude control algorithm can make the UAVs keep level flight on the desired altitude, and the UAVs are performing a stable flocking on the horizontal plane. The state variables of the UAVs are displayed in Fig. 5, including the velocities, the heading



(a) The 3D view of the flock.

(b) The plan view onto the horizontal plane.

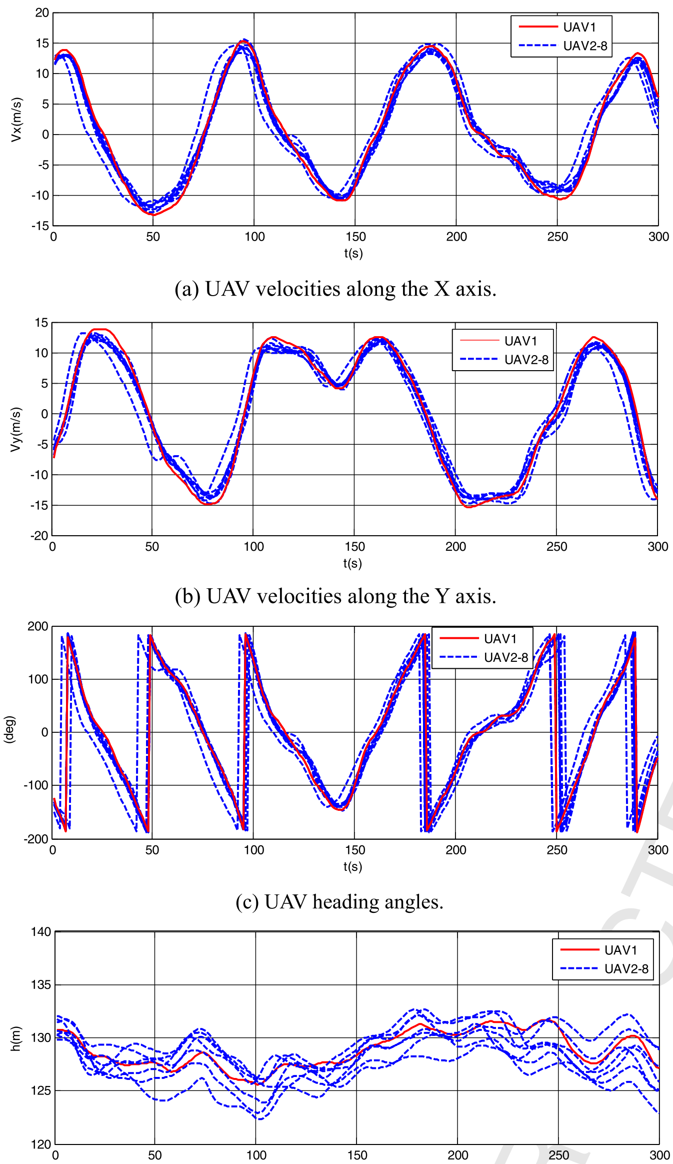
Fig. 4. Trajectories of the UAVs in performing a flock. (For interpretation of the colors in this figure, the reader is referred to the web version of this article.)

angles, and the altitudes. With the altitude consensus algorithm, UAVs maintain the flock at nearly the same altitude (130 m). In all the above simulation results, all UAVs satisfy their dynamic constraints. According to the proposed leader-follower interaction rules, the UAV flock keeps a stable formation on the same altitude without collision.

In order to illustrate the robustness of hierarchical leadership network, another simulation is conducted, in which a middle-level leader (set as UAV5) suddenly disappears and all the interactions related to it are cut off. The trajectories of the UAV flock in this simulation are presented in Fig. 6, in which the black cross indicates the position where UAV5 disappears. The corresponding state responses are shown in Fig. 7.

The simulation results demonstrate that pigeon hierarchical leadership network has advantages over the general network models when applying to the hierarchical multiple UAVs flocking problem. The proposed control framework uses the distributed control to solve the standard flocking problem in which the dynamics of all individuals are the same and of low dimension. It has been investigated that the convergence criterion depends on the full spectrum of the communication graph. For the hierarchical leadership communication topology of pigeon flock, which is directly connected graph and leader-following network, the consensus criterion is that the network has a directed spanning tree [24]. Further investigating the hierarchical network, we find that for a pigeon flock of N individuals, there exists at least $2N$ directed spanning trees. With these alternate connections, the hierarchical leadership

Please cite this article in press as: Q. Luo, H. Duan, Distributed UAV flocking control based on homing pigeon hierarchical strategies, Aerosp. Sci. Technol. (2017), http://dx.doi.org/10.1016/j.ast.2017.08.010



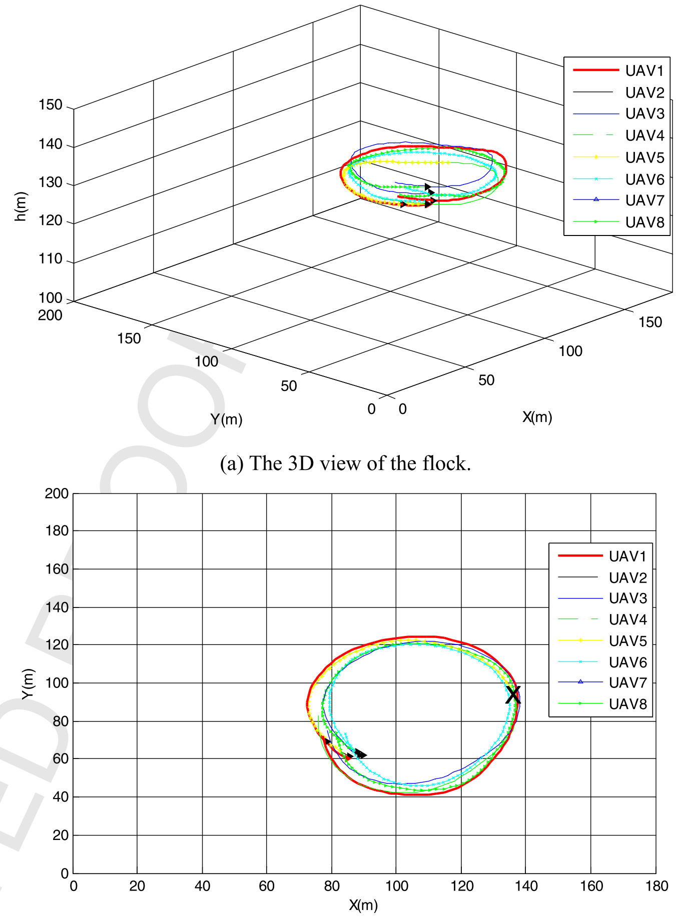
(a) UAV velocities along the X axis.

(b) UAV velocities along the Y axis.

(c) UAV heading angles.

(d) UAV altitudes.

Fig. 5. The time response of UAV velocity V_x and V_y, heading angle χ, and altitude h when performing a flock using the proposed algorithms.



(a) The 3D view of the flock.

(b) The plan view onto the horizontal plane.

Fig. 6. Trajectories of the UAVs in connection failure situations. (For interpretation of the colors in this figure, the reader is referred to the web version of this article.)

with pairwise leader-follower interaction can solve the connection failure problem without serious oscillations and avoid the local individual lost. Compared with the general multiple UAVs formation control method, the collective behavior inspired flocking algorithm does not need to construct the complicated control laws, and has better extensibility.

We also use fixed circle obstacles in an area (6×20) to compare the performance of the artificial physics (AP) and pigeon leadership hierarchies inspired methods in collision and obstacle avoidance. The artificial physics method is widely used to design and build rapidly deployable, scalable, adaptive, cost-effective, and robust networks of autonomous distributed multi-agents due to its efficient mathematical analysis and simplicity [25]. This environment represents a complex, unknown, crowded two-dimensional environment.

In the AP method, by using the Newtonian gravitational force law, the attractive force and repulsive force on UAVs can be defined as [26]:

$$\boldsymbol{F}_{att} = \frac{G_{goal}}{\|p_{agent} - p_{goal}\|^2} \cdot \hat{r}_{agent \to goal} \tag{16}$$

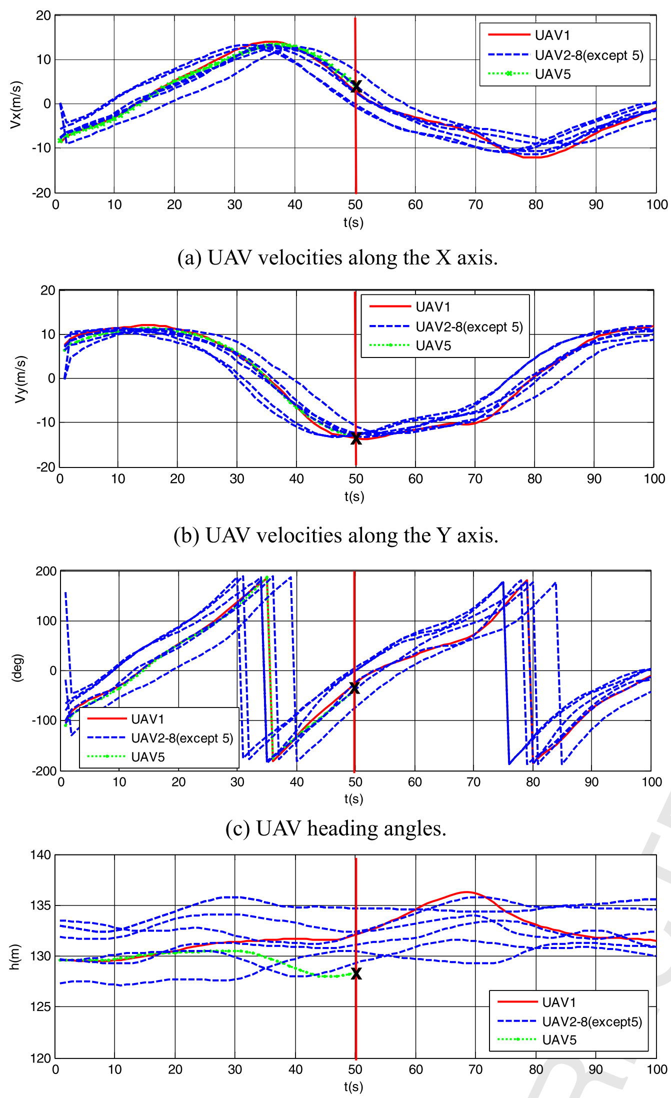
$$\boldsymbol{F}_{rep} = \begin{cases} \frac{G_{obs}}{\|p_{agent} - p_{obs}\|^2} \cdot \hat{r}_{obs \to agent} & \|p_{agent} - p_{obs}\| \le R_{rep} \\ 0 & \text{otherwise} \end{cases} \tag{17}$$

where $\|p_{agent} - p_{goal}\|$ represents the Euclidean distance between UAV and its goal, and $\hat{r}_{agent \to goal}$ represents the unit vector from p_{agent} to p_{goal}. G_{goal} is the gravitational constant of the goal. $\hat{r}_{obs \to agent}$ represents the unit vector from the obstacle to UAVs. R_{rep} is the action range of the repulsive force.

Fig. 8(a) and (b) show the trajectories of the UAVs generated by the AP method and pigeon inspired method, respectively. In Fig. 8 the red markers "•" indicate the obstacles. Both the two methods can reach the global destination through the complex obstacle area. By using the pigeon inspired method, the UAV flock spends much fewer time steps to get through the obstacle area. On the contrary, we can see that a large number of oscillations are caused by the AP method and the integrity of the flock can't be maintained.

Based on these experimental results and analysis, it is obvious that the proposed pigeon inspired framework has advantages over the AP method when applying to the UAV flocking in a complex environment. The pigeon leadership hierarchies based distributed control method achieves smoother trajectories without serious oscillations. Compared with the general AP method, the pigeon inspired framework does not need to construct the complicated potential function and establish virtual physics forces on the UAV.

Please cite this article in press as: Q. Luo, H. Duan, Distributed UAV flocking control based on homing pigeon hierarchical strategies, Aerosp. Sci. Technol. (2017), http://dx.doi.org/10.1016/j.ast.2017.08.010



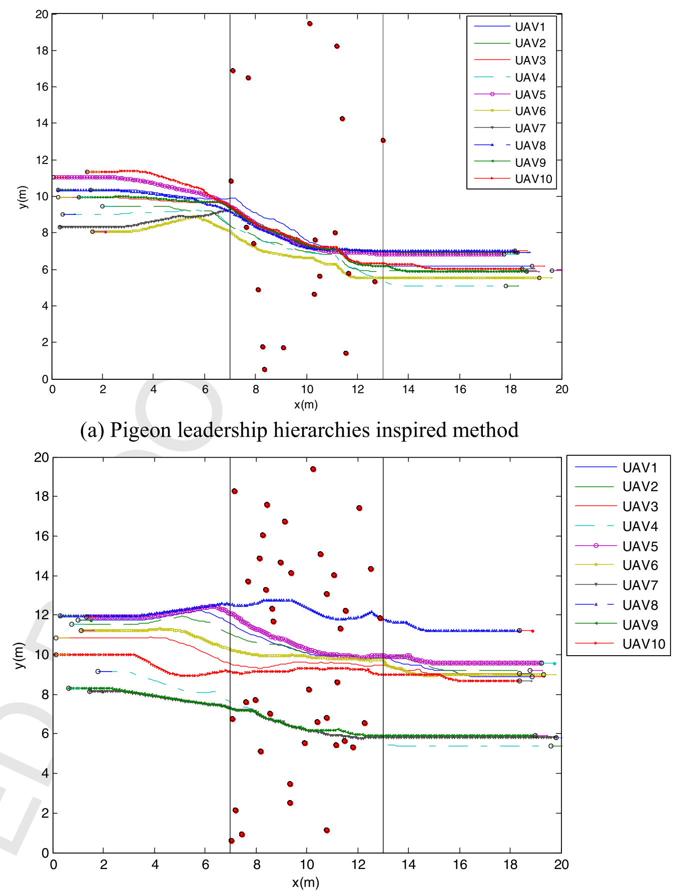
(a) UAV velocities along the X axis.

(b) UAV velocities along the Y axis.

(c) UAV heading angles.

(d) UAV altitudes.

Fig. 7. The time response of UAV velocity V_x and V_y, heading angle χ, and altitude h under connection failure.



(a) Pigeon leadership hierarchies inspired method

(b) Artificial physics method

Fig. 8. Trajectories of the UAVs with multiple circle obstacles using the two methods. (For interpretation of the colors in this figure, the reader is referred to the web version of this article.)

6. Conclusions

In this paper, a distributed control framework is proposed to solve the UAV flocking problem. The framework contains various hierarchical strategies based on the pigeon leadership hierarchies. Velocity correlation, leader-follower interaction, and hierarchical network are adopted to construct a stable flocking structure. System-specific features such as inertia, time delay, and general noise are considered in the flocking control. An altitude consensus algorithm is also employed to control all UAVs to fly in the same horizontal plane. Several simulations are conducted to verify the effectiveness and applicability of the proposed control framework. The distributed UAV flocking control framework is compared with the general AP method. The robustness of the proposed framework is also demonstrated. The simulation results indicate that the UAV flock is capable of keeping a stable flocking formation. Since the rapid and fierce movement of the fix-wing UAV is a challenge for collective behavior based control algorithms, future research should focus on improving the robustness of the flocking control system in both hardware and software. Furthermore, better solutions may be found by investigating the close connection between groups of organisms, and bio-inspired computation methods are also promising intelligent approaches for solving such type of complicated problems.

Conflict of interest statement

None declared.

Acknowledgements

This work was partially supported by National Natural Science Foundation of China under grant #61425008, #91648205, #61333004, Aeronautical Foundation of China under grant #2015ZA51013, and Innovation Foundation of BUAA for PhD Graduates. The authors would like to thank the editors and reviewers for their critical review of this manuscript.

References

[1] A.S. Clare, M.L. Cummings, J.P. How, A.K. Whitten, O. Toupet, Operator objective function guidance for a real-time unmanned vehicle scheduling algorithm, J. Aerosp. Comput. Inf. Commun. 9 (4) (2012) 161–173.
[2] S. Zhao, Z. Hu, M. Yin, K.Z.Y. Ang, P. Liu, F. Wang, X. Dong, F. Lin, B.M. Chen, T.H. Lee, A robust real-time vision system for autonomous cargo transfer by an unmanned helicopter, IEEE Trans. Ind. Electron. 62 (2) (2015) 1210–1219.
[3] H.B. Duan, Q.N. Luo, G.J. Ma, Y.H. Shi, Hybrid particle swarm optimization and genetic algorithm for multi-UAVs formation reconfiguration, IEEE Comput. Intell. Mag. 8 (3) (2013) 16–27.
[4] N. Nigam, S. Bieniawski, I. Kroo, J. Vian, Control of multiple UAVs for persistent surveillance: algorithm and flight test results, IEEE Trans. Control Syst. Technol. 20 (5) (2012) 1236–1251.
[5] F. Giulietti, M. Innocenti, M. Napolitano, L. Pollini, Dynamic and control issues of formation flight, Aerosp. Sci. Technol. 9 (1) (2005) 65–71.

[6] W. Lin, Distributed UAV formation control using differential game approach, Aerosp. Sci. Technol. 35 (2014) 54–62.
[7] J.R. Riehl, G.E. Collins, J.P. Hespanha, Cooperative search by UAV teams: a model predictive approach using dynamic graphs, IEEE Trans. Aerosp. Electron. Syst. 47 (4) (2011) 2637–2656.
[8] N. Nigam, S. Bieniawski, I. Kroo, J. Vian, Control of multiple UAVs for persistent surveillance: algorithm and flight test results, IEEE Trans. Control Syst. Technol. 20 (5) (2012) 1236–1251.
[9] G.A. Garcia, S.S. Keshmiri, Biologically inspired trajectory generation for swarming UAVs using topological distances, Aerosp. Sci. Technol. 54 (2016) 312–319.
[10] N.W.F. Bode, D.W. Franks, A.J. Wood, Limited interactions in flocks: relating model simulations to empirical data, J. R. Soc. Interface 8 (2011) 301–304.
[11] M. Nagy, Z. Akos, D. Biro, D.T. Vicsek, Hierarchical group dynamics in pigeon flocks, Nature 464 (7290) (2010) 890–893.
[12] B. Pettit, A. Flack, R. Freeman, T. Guilford, D. Biro, Not just passengers: pigeons, Columba livia, can learn homing routes while flying with a more experienced conspecific, Proc. Biol. Sci. (2012) 20122160.
[13] B. Pettit, Z. Ákos, T. Vicsek, D. Biro, Speed determines leadership and leadership determines learning during pigeon flocking, Curr. Biol. 25 (23) (2015) 3132–3137.
[14] G. Vásárhelyi, C. Virágh, G. Somorjai, N. Tarcai, T. Szörényi, T. Nepusz, T. Vicsek, Outdoor flocking and formation flight with autonomous aerial robots, in: 2014 IEEE/RSJ Int. Conf. on Intell. Robot. Syst, 2014, pp. 3866–3873.
[15] S. Hauert, S. Leven, M. Varga, F. Ruini, A. Cangelosi, J.C. Zufferey, D. Floreano, Reynolds flocking in reality with fixed-wing robots: communication range vs. maximum turning rate, in: 2011 IEEE/RSJ Int. Conf. Intell. Robot. Syst, 2011, pp. 5015–5020.
[16] A. Kushleyev, D. Mellinger, C. Powers, V. Kumar, Towards a swarm of agile micro quadrotors, Auton. Robots 35 (4) (2013) 287–300.
[17] A. Bürkle, F. Segor, M. Kollmann, Towards autonomous micro UAV swarms, J. Intell. Robot. Syst. 61 (1–4) (2011) 339–353.
[18] A. Flack, Z. Ákos, M. Nagy, T. Vicsek, D. Biro, Robustness of flight leadership relations in pigeons, Anim. Behav. 86 (4) (2013) 723–732.
[19] I.D. Couzin, J. Krause, N.R. Franks, S.A. Levin, Effective leadership and decision-making in animal groups on the move, Nature 433 (7025) (2005) 513–516.
[20] B. Pettit, A. Perna, D. Biro, D.J.T. Sumpter, Interaction rules underlying group decisions in homing pigeons, J. R. Soc. Interface 10 (89) (2013) 20130529.
[21] A. Flack, R. Freeman, T. Guilford, D. Biro, Pairs of pigeons act as behavioural units during route learning and co-navigational leadership conflicts, J. Exp. Biol. 216 (8) (2013) 1434–1438.
[22] Y. Huang, H. Wang, P. Yao, Energy-optimal path planning for Solar-powered UAV with tracking moving ground target, Aerosp. Sci. Technol. 53 (2016) 241–251.
[23] J. Wang, M. Xin, Integrated optimal formation control of multiple unmanned aerial vehicles, IEEE Trans. Control Syst. Technol. 21 (5) (2013) 1731–1744.
[24] H. Kim, H. Shim, J.H. Seo, Output consensus of heterogeneous uncertain linear multi-agent systems, IEEE Trans. Autom. Control 56 (1) (2011) 200–206.
[25] W.M. Spears, D.F. Spears, J.C. Hamann, et al., Distributed, physics-based control of swarms of vehicles, Auton. Robots 17 (2–3) (2004) 137–162.
[26] R. Olfati-Saber, Flocking for multi-agent dynamic systems: algorithms and theory, IEEE Trans. Autom. Control 51 (3) (2006) 401–420.